

22 A row of 12 decorative lights, connected in series, is connected to a mains supply. When the supply is switched on, the lights do not work. A voltmeter is used to test the circuit. When the voltmeter is connected across the third bulb in the row, a reading of zero is obtained.

Which of the following **cannot** be the **only** fault in the circuit?

- A** the filament of the third bulb has broken
- B** the filament of one of the other bulbs has broken
- C** the fuse in the mains supply has blown
- D** there is a break in the wire from the supply to the lights

23 A long straight wire XY lies in the same plane as a square loop of wire PQRS which is free